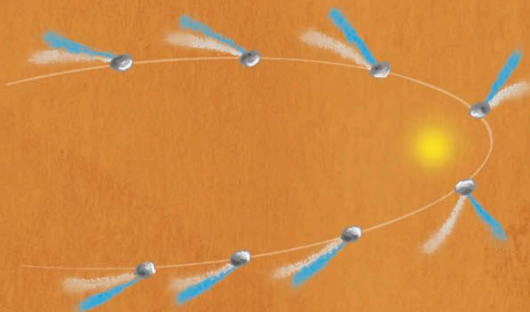


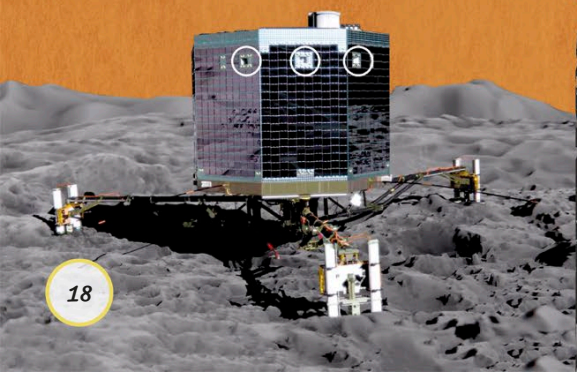
Impact crater Though it is rare, comets and meteorites can slam into the Earth with huge amounts of force. This leaves large impact craters, such as this one in Arizona. We even think a comet impact wiped out the dinosaurs!



Two tails A comet actually has two tails—a dust tail that stretches out behind it, and a gas tail that is pushed away from the sun by solar winds.



Rosetta The European space probe *Rosetta* reached Comet 67P in 2014. The probe released a lander called *Philae*, which was the first object to ever land on a comet and helped scientists learn a lot more about them.



Comet

Comets are frozen, dusty worlds created from material that was left over after our solar system formed. Some comets are so far away from the sun that we can't see them even with telescopes. However, many others have been spotted in the night sky.

Comets spend most of their time in an area called the Kuiper Belt at the edge of our solar system.

A comet can be pushed out of the Kuiper Belt after colliding with another comet.

Rocky beginnings

Far away from the sun, it is so cold that water and gases freeze, collecting on the surface of dust. These materials start to be pulled together by gravity, forming icy rocks that continue to combine until eventually, a huge icy, rocky comet is formed.